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| **Started on** | Wednesday, 7 October 2026, 10:27 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 7 October 2026, 10:42 PM |
| **Time taken** | 15 mins 54 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    scanf("%d",&arr[i]);
    int dp[n];
    for(int i=0;i<n;i++)
    dp[i]=1;
    for(int i=1;i<n;i++){
        for(int j=0;j<i;j++){
            if(arr[j]<=arr[i]){
                if(dp[j]+1>dp[i])
                dp[i]=dp[j]+1;
            }
        }
    }
    int ans=0;
    for(int i=0;i<n;i++){
        if(dp[i]>ans)
        ans=dp[i];
    }
    printf("%d",ans);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9<br>-1 3 4 5 2 2 2 2 3 | 6 | 6 | ✔ |
| ✔ | 7<br>1 2 2 4 5 7 6 | 6 | 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.